

AI Testing Basics - Quick Guide

A short reference for QA engineers transitioning into AI/GenAI testing.

What is AI Testing?

AI Testing validates AI/ML systems to ensure accuracy, reliability, fairness, safety, and performance.

Types of AI Systems

Machine Learning, Generative AI, NLP Chatbots, Computer Vision, Recommendation Systems.

Key Validation Areas

Accuracy, Hallucination Detection, Bias & Fairness, Security, Explainability, Performance, Robustness.

AI Testing Techniques

Functional Testing, Prompt Testing, Model Validation, Regression Testing, Adversarial Testing, A/B Testing.

GenAI Test Examples

Verify response relevance, factual correctness, safety compliance, response consistency, and prompt injection resistance.

Important Metrics

Precision, Recall, F1 Score, Accuracy, Latency, Response Quality, Toxicity Score.

AI Testing Challenges

Non-deterministic outputs, changing models, data quality issues, hallucinations, and bias.

Tools

Jupyter, Python, Promptfoo, DeepEval, LangSmith, Azure AI Evaluation, Postman.

QA Engineer Learning Path

Learn Python → AI Fundamentals → Prompt Engineering → LLM Testing → AI Test Automation.

Interview Tip

Focus on AI risks, evaluation metrics, hallucination testing, bias testing, and prompt validation.

Prepared as a concise AI Testing reference.